

.....💣💻 Data Science 💻💣.....

Defination of Data Science

In []: It is process of extracting insights data by using "Scientific Methods"

In []: Data Science is a branch of computer science where we study how to store, use and analyse data for deriving information from it.

In []: "Data science" is a deep study of the massive amount of data, which involves extracting meaningful insights from raw, structured, and unstructured data that is processed using the scientific method, different technologies, and algorithms.

Scientific Methods

In []:

- 1.Machine Learning
- 2.Deep Learning
- 3.Natural Language Processing(NLP)
- 4.Statistics
- 5.Visualization Tools (Seaborn, Matplotlib)

Type of Data

In []:

- 1.Structured >> CSV, Excel files
- 2.Unstructured >> Images, Videos, Audios, Text
- 3.Semi-Structured >> JSON, HTML

Real world Application of Data Science

In []:

- 1.Chatbot
- 2.face recognition (goole photos , facebook) >> vKYC
- 3.Digitization of any documents:
 - 1.Google vision
 - 2.Amazon textract
 - 3.Pytesseract
- 4.HoUSE PRIZE PREDICTION
- 5.weather Prection
- 6.Speech recognition (Alexa,Siri,Google Voice Assistant)
- 7.Google Search Engine
- 8.Google map
- 9.Sentiment Analysis
- 10.medical Diagnosis
- 11.Self driving Car (Tesla Cars)
- 12.Robotics

Machine Learning

<https://www.javatpoint.com/machine-learning-algorithms> (<https://www.javatpoint.com/machine-learning-algorithms>) (<https://www.javatpoint.com/machine-learning-algorithms>) (<https://www.javatpoint.com/machine-learning-algorithms>)

In []: Machine learning is program that analyse the data and learns to predict the outcome

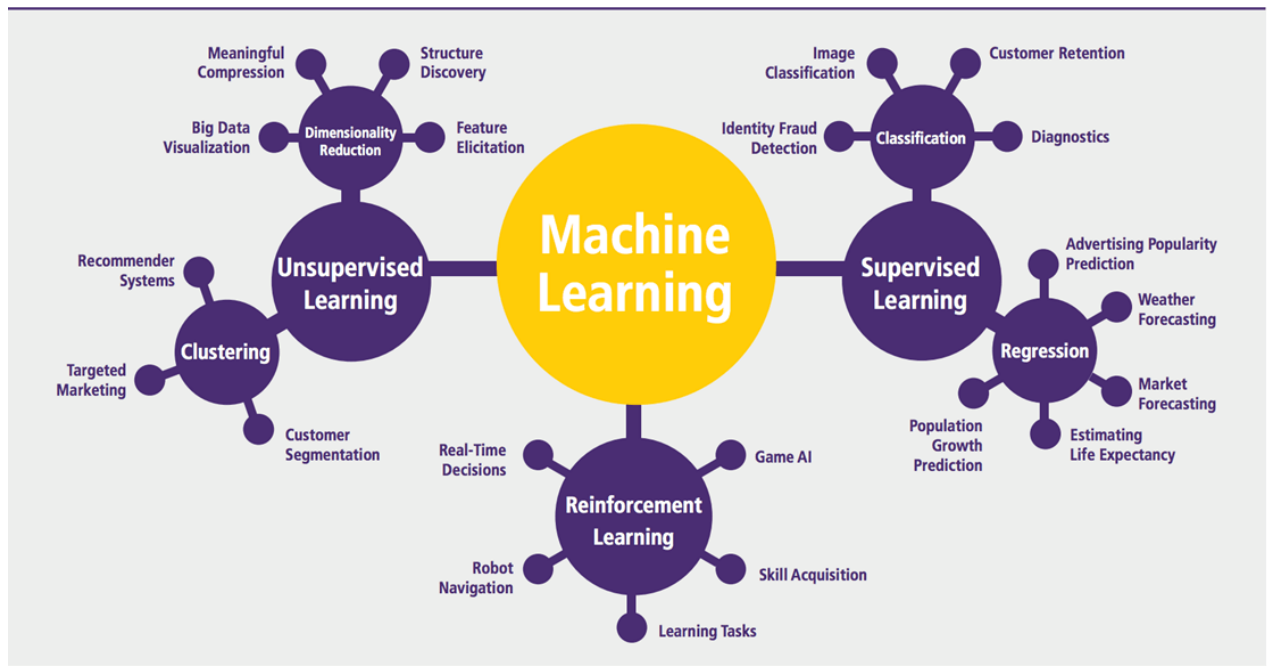
In []: "Algorithm" >> Set of instuctions that works on some given data in a particular way.

In []: In the real world, we are surrounded by humans who can learn everything from their experiences with their learning capability, and we have computers or machines which work on our instructions.
But can a machine also learn from experiences or past data like a human does? So here comes the role of Machine Learning.

"Machine learning" enables a machine to automatically learn from data, improve performance from experiences, and predict things without being explicitly programmed.

In []: Machine learning algorithms build a model based on sample data, known as training data, in order to make predictions or decisions without being explicitly programmed to do so.

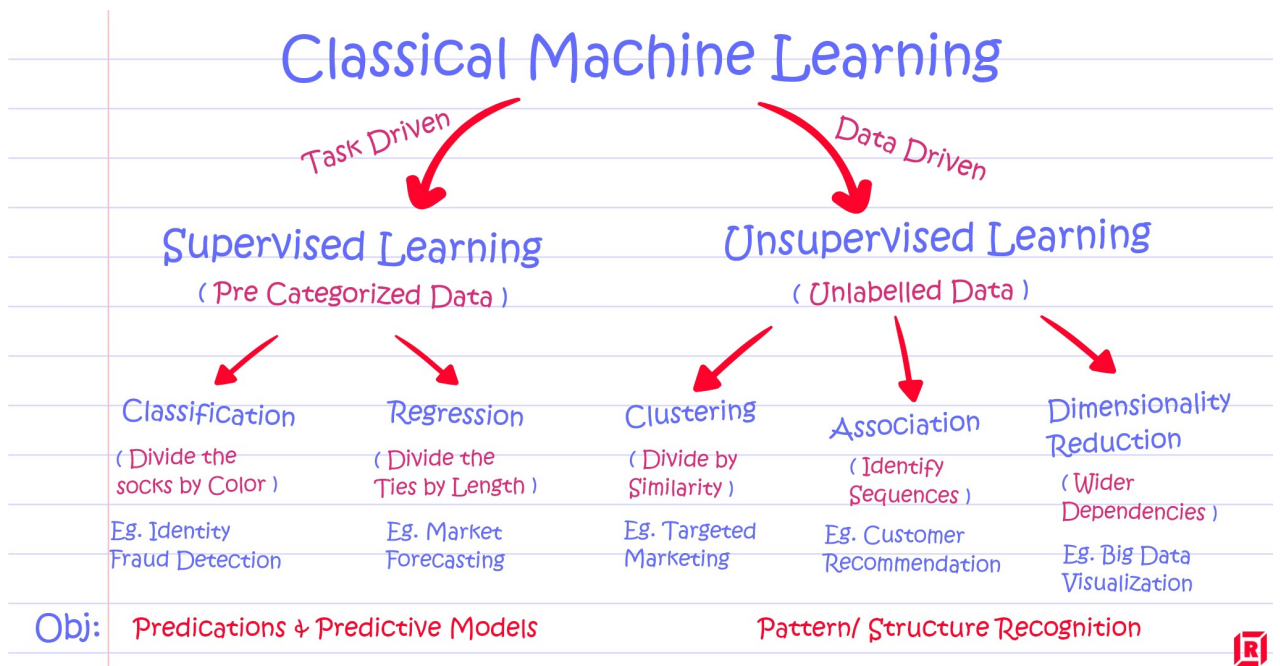
Machine learning algorithms are used in a wide variety of applications, such as in ""Medicine, Email Filtering, Speech Recognition, and Computer Vision"".



Types of machine Learning:

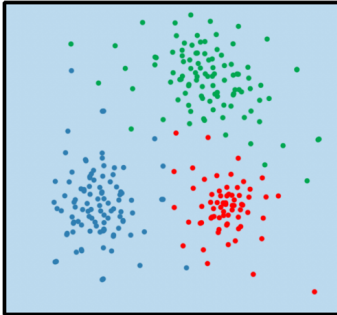
In []:

- i. Supervised Machine Learning
- ii. Unsupervised Machine Learning
- iii. Reinforcement Machine Learning

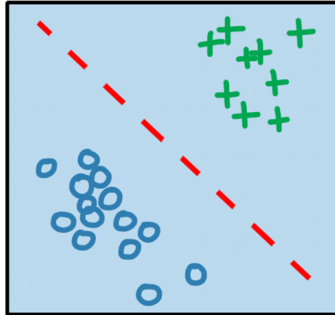


machine learning

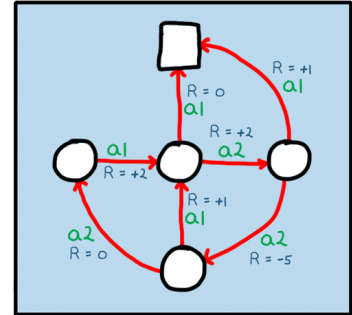
unsupervised
learning



supervised
learning



reinforcement
learning



i. Supervised Machine Learning

In []: Supervised ML uses labelled data (data with known answers i.e. We know Input & Outputs)

Supervised learning is the types of machine learning in which machines are trained using well "labelled" training data, and on basis of that data, machines predict the output.

The labelled data means some input data is already tagged with the correct output.

02 Types of Variables :

1. Independent Variables (Input Parameters, Predictions, Features, parameters)

>>> Values inside the column which is not dependent on values of another column.

An independent variable is the variable that is changed or controlled in a scientific experiment to test the effects on the dependent variable.

2. Dependent variable (Output, Target Variable)

>>> Values inside the column which is dependent on values of another column.

A dependent variable is the variable being tested and measured in a scientific experiment or calculation.

The dependent variable is 'dependent' on the independent variable.

As the experimenter changes the independent variable, the effect on the dependent variable is observed and recorded.

""if I change any value from any one column then then value from second column is change it is known as Dependent""

02 Types of Supervised Machine Learning :

1. "Classification" :(categorical data in target column)

Classification algorithms are used when the output variable is categorical, which means there are two classes such as Yes-No, Male-Female, True-false, etc.

1. Binary Classification (two categories in target column)

1/0, yes/no, True/False, Good/Bad, Pos/neg, Spam/not-Spam

2. Multiclass classification (More than 2 categories in target column)

High-medium-low,
Class0-class1-class2

for a classification algorithm that filters emails,

"Input is" would be an incoming email, and

"Output is" would be the name of the folder in which to file the email.

2. "Regression" : (Continuous Data in target column)

Regression algorithms are used if there is a relationship between the input variable and the output variable.

1. car prize

2. House prize

3. stock prize

"Applications"

- i. Recommendation Systems,
- ii. Visual Identity Tracking,
- iii. Face Verification,
- iv. Speaker Verification.
- v. Weather forecasting
- vi. Market Trends

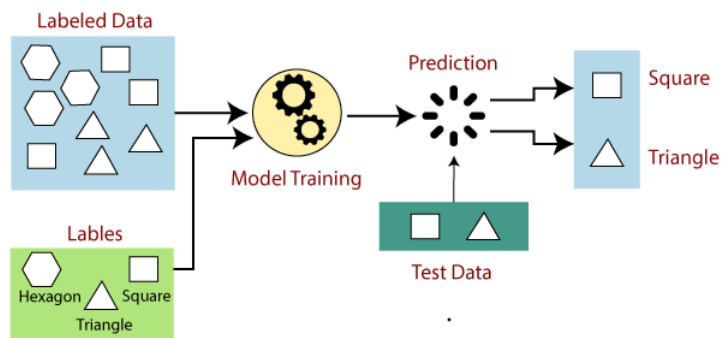
In []: Supervised learning algorithms build a mathematical model of a set of data that contains both the inputs and the desired outputs. This data is known as training data, and consists of a set of training examples.

Each training example has one or more inputs and the desired output, also known as a supervisory signal.

In the mathematical model, each training example is represented by an array or vector, sometimes called a feature vector, and the training data is represented by a `""matrix""`.

Steps Involved in Supervised Learning:

- In []:
- First Determine the type of training dataset
 - Collect/Gather the labelled training data.
 - Split the training dataset into training dataset, test dataset, and validation dataset.
 - Determine the input features of the training dataset, which should have enough knowledge so that the model can accurately predict the output.
 - Determine the suitable algorithm for the model, such as support vector machine, decision tree, etc.
 - Execute the algorithm on the training dataset. Sometimes we need validation sets as the control parameters, which are the subset of training datasets.
 - Evaluate the accuracy of the model by providing the test set. If the model predicts the correct output, which means our model is accurate.

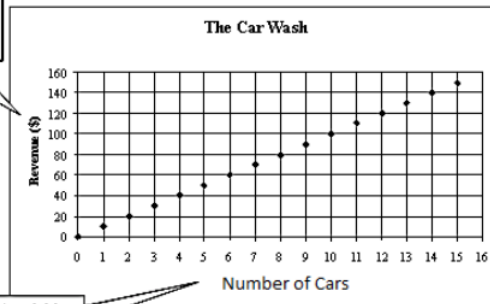


Independent Variables

Dependent Variable

Subject_ID	Age (years)	Weight (kg)	Systolic BP (mm Hg)
1	60	58	117
2	61	90	120
3	74	96	145
4	57	72	129
5	63	62	132
6	68	79	130
7	66	69	110
8	77	96	163
9	63	96	136
10	54	54	115
11	63	67	118
12	76	99	132
13	60	74	111
14	61	73	112
15	65	85	147
16	79	80	138

Dependent variable –
The amount of money raised **depends** on the number of cars washed.



Independent variable
- the number of cars washed

Supervised Machine Learning Algorithms:

In []:	1. Linear Rergession	Regression
	2. Logistic Rergession	Classification
	3. K-Nearest Neighbour	Classification and Regression
	3.1 KNNClassifier	
	3.2 KNNRegressor	
	4. Decision Tree	Classification and Regression
	5. Random Forest	Classification and Regression
	6. AdaBoost	Classification and Regression
	7. Gradient Boost	Classification and Regression
	8. XGBoost	Classification and Regression
	9. Support Vector Machine	Classification and Regression
	10. Naive Bayes Classifier	Classification

Advantages of Supervised learning:

- In []:
- i. With the help of supervised learning, the model can predict the output on the basis of prior experiences.
 - ii. In supervised learning, we can have an exact idea about the classes of objects.
 - iii. Supervised learning model helps us to solve various real-world problems such as fraud detection, spam filtering, etc.

Disadvantages of supervised learning:

- In []:
- i. Supervised learning models are not suitable for handling the complex tasks.
 - ii. In supervised learning, we cannot predict the correct output if the test data is different from the training dataset.
 - iii. Training required lots of computation times.
 - iv. In supervised learning, we need enough knowledge about the classes of object.

ii. Unsupervised Machine Learning

In []: If the data unlabelled , use Unsupervised ML

unlabelled >> No Dependent column , only independent variables (No any relation in between 1 value to the another value)

There may be many cases in which we do not have labeled data and need to find the hidden patterns from the given dataset. So, to solve such types of cases in machine learning, we need unsupervised learning techniques.

"Unsupervised learning" is a type of machine learning in which models are trained using unlabeled dataset and are allowed to act on that data without any supervision.

	Matches	Innings	Runs	wickets	(No any relation in between 1 value to the another value)
player1	100	95	5000	12	
player2	95	50	1000	78	
player3	110	75	3000	35	
.....					
player1000					

In []: Unsupervised learning algorithms take a set of data that contains only inputs, and find structure in the data, like grouping or clustering of data points.

The algorithms, therefore, learn from test data that has not been labeled, classified or categorized.

Instead of responding to feedback, unsupervised learning algorithms identify commonalities in the data and react based on the presence or absence of such commonalities in each new piece of data.

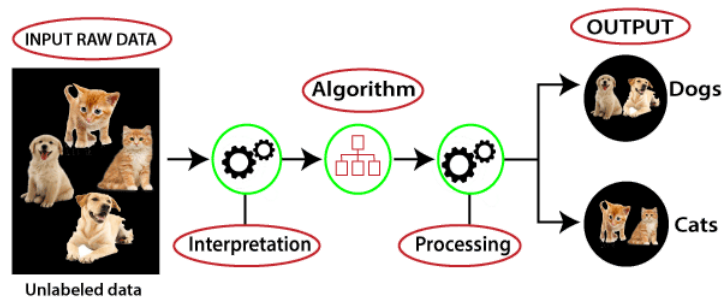
Though unsupervised learning encompasses other domains involving summarizing and explaining data features.

Unsupervised Machine Learning Algorithms:

```
In [ ]: 1. K-Means Clustering
        2. Hierarchical Clustering
        3. Principle component Analysis (PCA)
        4. KNN (k-nearest neighbors)
        5. Anomaly detection
        6. Neural Networks
        7. Independent Component Analysis
        8. Apriori algorithm
        9. Singular value decomposition
```

Why use Unsupervised Learning?

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In [ ]: i. Unsupervised learning is helpful for finding useful insights from the data.
        ii. Unsupervised learning is much similar as a human learns to think by their own experiences,
            which makes it closer to the real AI
        iii. Unsupervised learning works on unlabeled and uncategorized data which make unsupervised learning more important.
        iv. In real-world, we do not always have input data with the corresponding output so to solve such cases,
            we need unsupervised learning
```



Types of Unsupervised Learning Algorithm:

A. Cluster B. Association

A. Clusters

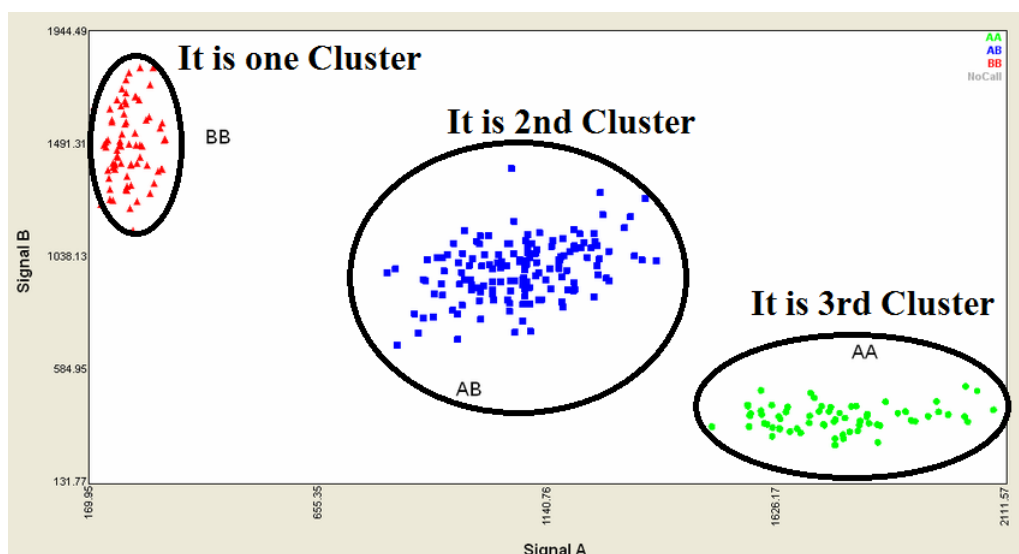
```
In [ ]: It is basically a type of unsupervised learning method.
```

Unsupervised learning algorithms take a set of data that contains only inputs, and find structure in the data, like grouping or clustering of data points.

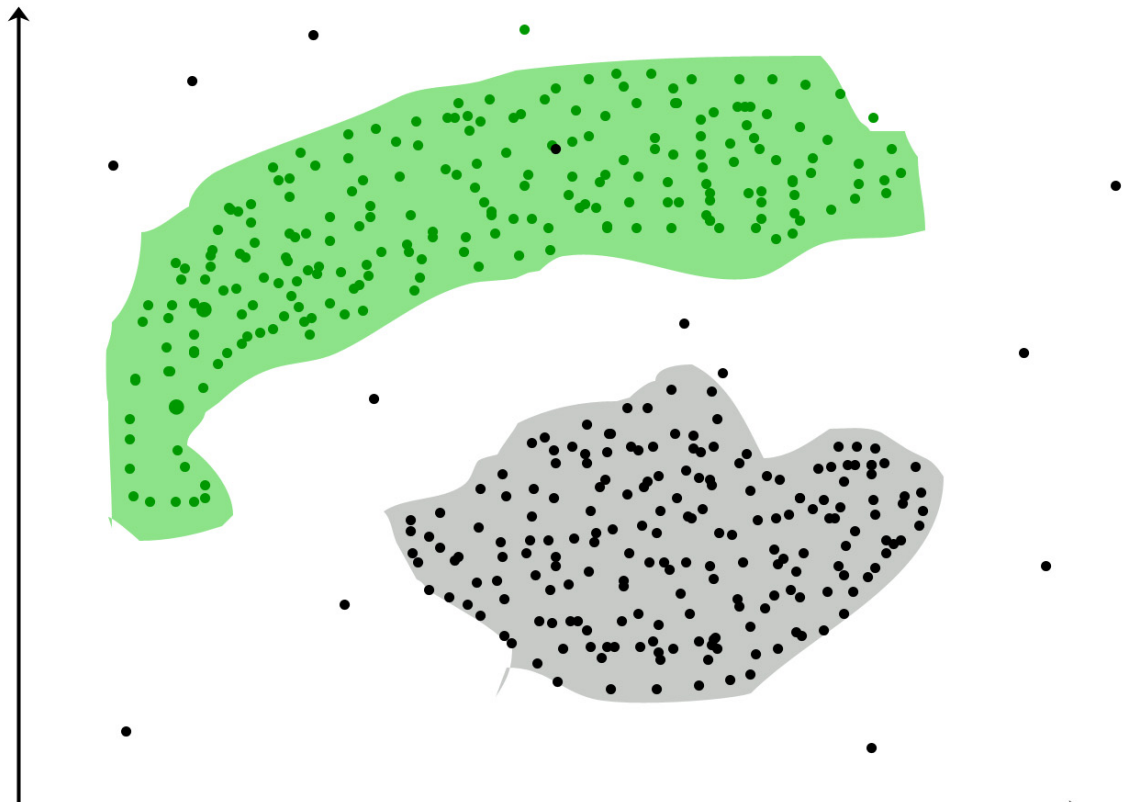
Clustering is the task of dividing the population or data points into a number of groups such that data points in the same groups are more similar to other data points in the same group and dissimilar to the data points in other groups.

It is basically a collection of objects on the basis of similarity and dissimilarity between them.

Clustering is very much important as it determines the intrinsic(belonging naturally)grouping among the unlabelled data present. There are no criteria for good clustering. It depends on the user, what is the criteria they may use which satisfy their need.



It is not necessary for clusters to be spherical. Such as :



B. Association

In []: An association rule **is** an unsupervised learning method which **is** used **for** finding the relationships between variables **in** the large database.

It determines the **set** of items that occurs together **in** the dataset.

Association rule makes marketing strategy more effective.

Such **as** people who buy X item (suppose a bread) are also tend to purchase Y (Butter/Jam) item.

A typical example of Association rule **is** Market Basket Analysis.

Advantages of Unsupervised Learning

- In []:
- i. Unsupervised learning **is** used **for** more **complex** tasks **as** compared to supervised learning because, **in** unsupervised learning, we don't have **labeled input data**.
 - ii. Unsupervised learning **is** preferable **as** it **is** easy to get unlabeled data **in** comparison to labeled data.

Disadvantages of Unsupervised Learning

- In []:
- i. Unsupervised learning **is** intrinsically more difficult than supervised learning **as** it does **not** have corresponding output.
 - ii. The result of the unsupervised learning algorithm might be less accurate **as** **input** data **is not** labeled, **and** algorithms do **not** know the exact output **in** advance.

Supervised Learning	Unsupervised Learning
Supervised learning algorithms are trained using labeled data.	Unsupervised learning algorithms are trained using unlabeled data.
Supervised learning model takes direct feedback to check if it is predicting correct output or not.	Unsupervised learning model does not take any feedback.
Supervised learning model predicts the output.	Unsupervised learning model finds the hidden patterns in data.
In supervised learning, input data is provided to the model along with the output.	In unsupervised learning, only input data is provided to the model.
The goal of supervised learning is to train the model so that it can predict the output when it is given new data.	The goal of unsupervised learning is to find the hidden patterns and useful insights from the unknown dataset.
Supervised learning needs supervision to train the model.	Unsupervised learning does not need any supervision to train the model.
Supervised learning can be categorized in Classification and Regression problems.	Unsupervised Learning can be classified in Clustering and Associations problems.
Supervised learning can be used for those cases where we know the input as well as corresponding outputs.	Unsupervised learning can be used for those cases where we have only input data and no corresponding output data.
Supervised learning model produces an accurate result.	Unsupervised learning model may give less accurate result as compared to supervised learning.
Supervised learning is not close to true Artificial intelligence as in this, we first train the model for each data, and then only it can predict the correct output.	Unsupervised learning is more close to the true Artificial Intelligence as it learns similarly as a child learns daily routine things by his experiences.
It includes various algorithms such as Linear Regression, Logistic Regression, Support Vector Machine, Multi-class Classification, Decision tree, Bayesian Logic, etc.	It includes various algorithms such as Clustering, KNN, and Apriori algorithm.

When Should you Choose Supervised Learning vs. Unsupervised Learning?

In []: In manufacturing, a large number of factors affect which machine learning approach is best for any given task. And, since every machine learning problem is different, deciding on which technique to use is a complex process. In general, a good strategy for honing in on the right machine learning approach is to:

1. "Evaluate The Data":
Is it labeled/unlabelled? Is there available expert knowledge to support additional labeling?
This will help to determine whether a supervised, unsupervised, semi-supervised or reinforced learning approach should be used
2. "Define The Goal":
Is the problem recurring, defined one? Or, will the algorithm be expected to predict new problems?
3. "Review Available Algorithms":
That may suit the problem with regards to dimensionality (number of features, attributes or characteristics).
Candidate algorithms should be suited to the overall volume of data and its structure
4. "Study Successful Applications" of the algorithm type on similar problems

3.Reinforcement Learning

In []: Reward based Learning

"Reinforcement learning algorithms are used in"

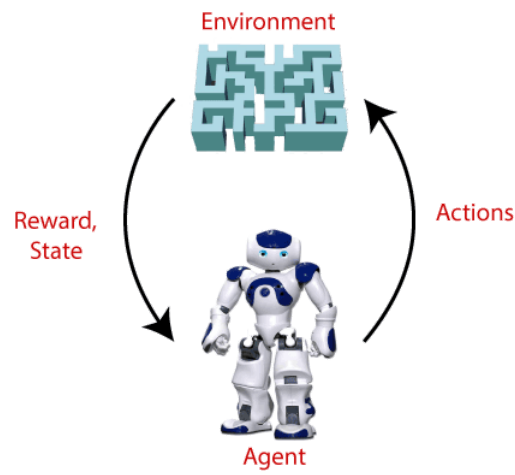
1. Self driving cars
2. Temperature control system
3. Learning to play a game against a human opponent.

In []: Reinforcement Learning is a feedback-based Machine learning technique in which an agent learns to behave in an environment by performing the actions and seeing the results of actions.
For each good action, the agent gets positive feedback, and for each bad action, the agent gets negative feedback or penalty.

In Reinforcement Learning, the agent learns automatically using feedbacks without any labeled data, unlike supervised learning. Since there is no labeled data, so the agent is bound to learn by its experience only.
RL solves a specific type of problem where decision making is sequential, and the goal is long-term, such as game-playing, robotics, etc.

"Reinforcement learning" is a type of machine learning method where an intelligent agent (computer program) interacts with the environment and learns to act within that

<https://www.javatpoint.com/reinforcement-learning>



Reinforcement Learning	Supervised Learning
RL works by interacting with the environment.	Supervised learning works on the existing dataset.
The RL algorithm works like the human brain works when making some decisions.	Supervised Learning works as when a human learns things in the supervision of a guide.
There is no labeled dataset is present	The labeled dataset is present.
No previous training is provided to the learning agent.	Training is provided to the algorithm so that it can predict the output.
RL helps to take decisions sequentially.	In Supervised learning, decisions are made when input is given.

Regression Analysis:

In []: Regression analysis is a statistical method to model the relationship between a dependent (target) and independent (predictor) variables with one or more independent variables.

Regression shows a line or curve that passes through all the datapoints on target-predictor graph in such a way that the vertical distance between the datapoints and the regression line is minimum.

In []: Types of Regression:

1. Linear Regression
2. Logistic Regression
3. Polynomial Regression
4. Support Vector Regression
5. Decision Tree Regression
6. Random Forest Regression
7. Ridge Regression
8. Lasso Regression: